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The effect of plyometric training on distance running performance

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Abstract Previous research has reported that plyometric training improves running economy (RE) and ultimately distance-running performance, although the exact mechanism by which this occurs remains unclear. This study examined whether changes in running performance resulting from plyometric training were related to alterations in lower leg musculotendinous stiffness (MTS). Seventeen male runners were pre- and post-tested for lower leg MTS, maximum isometric force, rate of force development, 5-bound distance test (5BT), counter movement jump (CMJ) height, RE, $\dot{V}O_{2\max}$, lactate threshold (Th_{la}), and 3-km time. Subjects were randomly split into an experimental (E) group which completed 6 weeks of plyometric training in conjunction with their normal running training, and a control (C) group which trained as normal. Following the training period, the E group significantly improved 3-km performance (2.7%) and RE at each of the tested velocities, while no changes in $\dot{V}O_{2\max}$ or Th_{la} were recorded. CMJ height, 5BT, and MTS also increased significantly. No significant changes were observed in any measures for the C group. The results clearly demonstrated that a 6-week plyometric programme led to improvements in 3-km running performance. It is postulated that the increase in MTS resulted in improved RE. We speculate that the improved RE led to changes in 3-km running performance, as there were no corresponding alterations in $\dot{V}O_{2\max}$ or Th_{la} .

Keywords Plyometrics · Running economy · Musculotendinous stiffness · Distance running

Introduction

Whilst $\dot{V}O_{2\max}$ has previously been well regarded as a good predictor of endurance performance, other variables such as maximal aerobic velocity and running economy (RE) have been cited as better predictors of distance running performance in trained athletes (Morgan et al. 1989; Paavolainen et al. 1999). RE is simply defined as the steady-state oxygen requirement for a given submaximal running velocity (Conley et al. 1981; Morgan et al. 1989). Numerous factors have been associated with RE, including heart rate, ventilation, $\dot{V}O_{2\max}$, gender, age, temperature, fatigue, training status, body and segmental mass distribution, stride length, and other biomechanical variables (Daniels 1985; Morgan et al. 1989; Pate et al. 1989; Bailey and Pate 1991). Bailey and Pate (1991) noted that while various factors have been found to affect RE, relatively little research has been conducted into examining potential avenues for improving RE by means of manipulating the aforementioned factors.

Traditionally, the favoured training method of improving RE was to perform long slow distance training, known as LSD (Temple 1990). Sjodin and Svedenhag (1985) reported that the more training an athlete does, up to a cut off point of 120 km week⁻¹, the more efficient the athlete will be. However, Noakes (1991) noted that whilst high weekly training distances may be a factor in increasing RE, the associated risks with such mileage were high. Additionally, it has been demonstrated that high intensive interval training such as 400 m through to 1 mile repetitions can lead to improvements in RE (Conley et al. 1981, 1984; Sjodin and Svedenhag 1982; Daniels 1998). Thus it is apparent that the optimal training method for improving RE remains to be determined.

Recent research has focussed on the effects of various forms of strength training on RE and running performance. Johnston et al. (1997) examined the effects of a 10-week strength training programme on female

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distance runners. It was reported that such a programme when conducted in conjunction with endurance training significantly improved RE (4%). Paavolainen et al. (1999) examined the effects of explosive-strength (plyometric) training on 5-km running time, RE, and muscle power. The authors concluded that simultaneous explosive-strength and endurance training produced a significant improvement in 5-km running performance in well-trained endurance athletes. The authors suggested that the improvements were due to improved neuromuscular characteristics that were transferred into enhanced muscle power and RE.

Interestingly, Noakes (1991) speculated that runners with poor economy might have muscles that are less able to utilise the impact energy produced as they eccentrically absorb the force of landing. It has been further suggested that the relative stiffness of the musculotendinous system (MTS) may determine the body's ability to store and utilise such energy (Wilson et al. 1991). Cornu et al. (1997) noted that there are a number of structures that are potentially modifiable as a result of plyometric training including the contractile component, the series elastic component (SEC) and the parallel elastic component. The authors reported that 7 weeks of plyometric training led to increases in tendon stiffness. Recent research has further provided evidence supporting a conceptual link between MTS and RE. Dalleau et al. (1998) noted that the energy cost of running was significantly related to the stiffness of the propulsive leg and that when the apparent stiffness of a subject decreased, the energy cost of running subsequently increased. Similarly, Heise and Martin (1998) concluded that less economical runners may possess a more compliant running style during ground contact. It was therefore hypothesised in the current study that plyometric training would increase MTS and would lead to improvements in RE through enhanced utilisation of stored impact energy during running. We proposed that due to the increase in stiffness derived from the plyometric training, subjects would achieve greater propulsion for the same or less

energy cost, thereby improving economy and consequently running performance.

Methods

Subjects

Seventeen male distance runners [mean (SD)] (age: 25 (4) years, mass: 72.4 (5.5) kg, height: 178 (4) cm) with an average training history of 10 (6) years were randomly assigned to experimental (E) ($n=8$) and control (C) groups ($n=9$). All subjects (E and C groups) had an average weekly running regime of approximately 60–80 km. As running training was not manipulated in the current study, subjects in both groups continued with their normal running over the course of the experiment. Training logs for all subjects were monitored prior to and during the training period to ensure that running volumes and intensities were not altered. It was a requirement of the study that participants had not previously undertaken a structured plyometric training programme and had not performed plyometric exercises in the previous 3 months. The study was approved by the Human Ethics Committee of the University of Technology, Sydney, and all participants were fully informed of the nature of the testing and training as well as the associated risks and signed written consent forms prior to the commencement of the study.

Plyometric training

After pre-testing, the E group completed 6 weeks of plyometric training. The plyometric programme involved two sessions per week for the first 3 weeks and three sessions per week for the final 3 weeks. Exercises consisted of various jumps, bounds, and hops in both horizontal and vertical planes. Prior to each session, all subjects underwent a 20-min dynamic warm up including leg swings, ankle bounces, skips, run throughs, as well as static stretching. Progressive overload principles were incorporated into the programme by increasing the number of foot contacts and varying the complexity of the exercises (see Table 1). For all exercises subjects were instructed to give maximal efforts with minimal ground contact times.

Measurements

Both groups were tested prior to and after the 6-week training period for all variables. The pre- and post-testing consisted of two separate test sessions with a minimum of 72 h between each test

Table 1 Six-week plyometric training programme (e.g. 2×10 = two sets of ten repetitions)

Week/session	Squat jump	Split scissor jump	Double leg bound	Alternate leg bound	Single leg forward hop	Depth jump	Double leg hurdle jump	Single leg hurdle hop	Total contacts
1/1	2×10	2×10	2×10						60
1/2	2×10	2×10	2×10						60
2/1	2×10	2×10	2×10	2×10					100
2/2	2×10	2×10	2×10	2×10					100
3/1		2×12	2×12	2×12	2×10				136
3/2		2×12	2×12	2×12	2×10				136
4/1			3×10	3×10	2×12	2×6			150
4/2			3×10	3×10	2×12	2×6			150
4/3			3×10	2×15	3×10	2×8			156
5/1				2×15		2×8	2×10	2×10	136
5/2				3×15		2×10	2×10	2×10	170
5/3				3×15		2×10	2×10	2×10	170
6/1					3×10	3×10	3×10	3×10	180
6/2					2×15	3×10	3×10	3×10	180
6/3					2×15	3×10	3×10	3×10	180

day. The variables tested on day 1 included: RE, aerobic power ($\dot{V}O_{2\max}$), lactate threshold (Th_{la}), MTS of the lower leg, maximal isometric force ($F_{iso,\max}$), rate of force development (RFD). Day 2 testing consisted of a counter movement jump (CMJ) test, a 5-bound test (5BT), and a 3-km time trial.

RE, $\dot{V}O_{2\max}$, Th_{la} tests

Once resting data was recorded and subjects had completed a warm-up, RE, $\dot{V}O_{2\max}$, and Th_{la} were determined during an incremental treadmill test. Commencing at a velocity of 10 km h⁻¹ and a 1% gradient, the subject ran for 3 min. After 3 min the treadmill was stopped for 1 min in order to sample blood for the blood lactate analysis. After each 1-min blood sampling period, the treadmill was increased by increments of 2 km h⁻¹. The subjects ran each of these increments for 3 min, before the 1-min blood collection period. Once the treadmill reached a velocity of 20 km h⁻¹, an increase of 2% gradient was substituted every 3 min for further increases in velocity. The test continued until volitional fatigue, whereby the subject could no longer maintain the treadmill velocity. Expired gases were collected in a mixing chamber and measured continuously using a Morgan Benchmark Gas Analyser. Prior to each test the gas analyser was calibrated with gases of a known composition. Gas values were obtained in 10-s intervals, with the values collected from the last minute of each 3-min interval being averaged to determine the oxygen uptake for that running velocity. Since RE is defined as the oxygen demand for a standardised speed (Conley and Krahenbuhl 1980), the RE in this study was determined by averaging the $\dot{V}O_2$ values for the last minute at velocities of 12, 14, and 16 km h⁻¹. A YSI Biochemistry Analyser (model 2700 Select, John Morris Scientific, Chatswood, NSW, Australia) was used to analyse the blood lactate levels at the completion of each 3-min interval. A 30- μ l volume of blood was sampled from the left ear lobe of each of the subjects. Heart rate monitors (Polar, Finland) were used to record average values during the last 10 s of each 3-min interval.

$F_{iso,\max}$ and RFD tests

The $F_{iso,\max}$ and RFD of the triceps surae were measured independently for each leg by performing a maximal isometric contraction on an instrumented seated calf raise machine (Onspot P/L, Wollongong, Australia). The seated calf raise machine was engineered with a built-in 500 N load cell (Chase Engineering, Sydney, Australia) placed under a rigid footplate, which enabled measurement of force (See Fig 1). Adjustable arm and knee pads were used to accommodate different limb lengths. A multi-link chain allowed the arm to be fixed at the desired angle, in order to act as an anchor during the $F_{iso,\max}$ test ensuring a static contraction. Subjects sat in the apparatus without shoes, in an upright position with hips, knees, and ankles positioned at 90°. This position was selected as it isolated the triceps surae musculature and allowed testing to be conducted at an ankle angle, which was approximately that at which the highest impact of running occurs. Machine setting positions were recorded to ensure consistency during both pre- and post-tests. Arms were folded across the chest and the non-tested leg placed to the side in a straight position resting on the heel in order to eliminate any influence on the force production of the tested leg. Subjects were instructed to produce maximal force against the knee pad as fast as possible and to hold it for 3 s. The force data was sampled at 1,000 Hz and recorded to a computer. $F_{iso,\max}$ was calculated as the peak of the force curve and this data was used to calculate the relative loads for the MTS tests. RFD was determined as the derivative of the force curve using a 5-ms interval and the peak RFD value was recorded.

MTS test

The oscillation technique previously reported by several authors was used in the present study in order to determine MTS of the

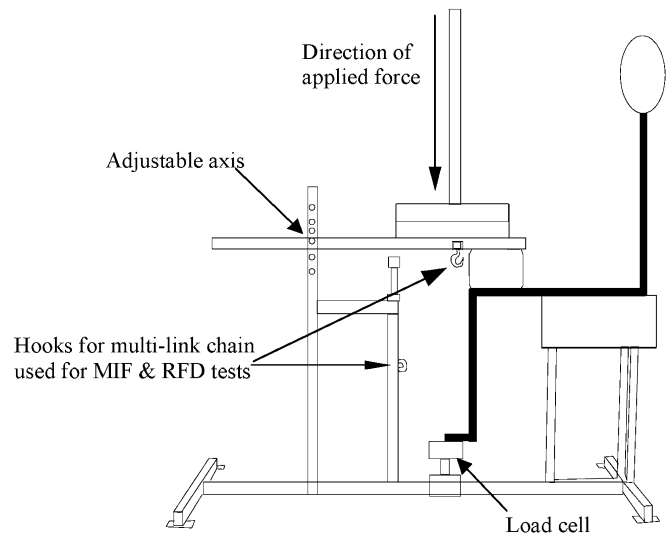


Fig. 1 Instrumented seated calf raise machine used in the study to assess unilateral maximal isometric force ($F_{iso,\max}$), rate of force development (RFD), and musculotendinous stiffness (MTS)

lower limb (Cavagna 1970; Shorten 1987; Wilson et al. 1991, 1992, 1994; McNair and Stanley 1996; Walshe et al. 1996; Walshe and Wilson 1997). Subjects were placed in the same position as they were for the $F_{iso,\max}$ and RFD tests; however the multi-link chain was removed. Loads of 25, 50, and 75% of the $F_{iso,\max}$ were supported by the subjects utilising the triceps surae with an ankle angle of 90°. The 75% load was deemed the maximal test load as subjects could not reliably maintain a steady position during the test procedure for greater loads. Additionally Wilson et al. (1991) reported that stiffness values obtained at approximately 70% of maximal load were equivalent to the maximal value of MTS. On average, this load represented approximately 1.4 times the body mass of subjects. Such a force is comparable to the vertical impact force experienced by subjects during submaximal running (Williams and Cavanagh 1987).

A brief perturbation in the order of 100 N to 200 N was applied by the experimenter to the machine arm on which the load was placed. Slight variations in the magnitude of the perturbation between trials does not effect the stiffness values as a spring system oscillates at its natural frequency independent of the magnitude of the applied perturbation (Shorten 1987; Wilson et al. 1991, 1992, 1994; Walshe et al. 1996; Walshe and Wilson 1997). During the period of perturbation, the subjects were instructed to retain the 90° ankle angle by maintaining a constant level of muscular activity. This was achieved by employing a 'do not intervene' strategy, which results in limiting the neural response to the applied perturbation to only those that are involuntary (Wilson et al. 1991). The oscillations were sampled by the load cell (1,000 Hz) located under the footplate and recorded on a computer. The data was filtered using a low pass Butterworth filter (4th order) with a cut-off frequency of 12 Hz. The resulting damped force oscillations were then used to calculate MTS as has been extensively described in the literature (Cavagna 1970; Shorten 1987; Wilson et al. 1991, 1992, 1994; McNair and Stanley 1996; Walshe et al. 1996; Walshe and Wilson 1997).

CMJ tests

After completing a standardised warm up, subjects performed three CMJ tests on a Swift Performance Jump Mat (Lismore, Australia). Subjects were instructed to jump for maximal height keeping their hands placed on their hips during the jump in order to reduce any contribution from the upper body. No restrictions were placed on the knee angle during the eccentric phase of the CMJ. Upon

completion of the three CMJ tests, the best jump height was recorded for analysis.

5-Bound test

The 5BT involved the subjects attempting to cover the greatest horizontal distance possible by performing a series of five forward jumps with alternate left and right foot contacts (Paavolainen et al. 1999). Subjects were given three trials, with the best result (greatest distance) recorded. This test was used in order to indicate performance improvements in the utilisation of the stretch shortening cycle (SSC) as a result of the plyometric training.

3-km time

Running performance was determined from a 3-km time trial conducted on an outdoor tartan athletic track. In order to ensure maximum effort, subjects were ranked on recent 3-km performance with the more able subjects competing in one time-trial, and the remaining participants competing in another time-trial. In order to minimise any 'pacing' or other 'race type' strategies affecting individual performances, all subjects were instructed to strive for their best 3-km run exerting a consistent effort over the distance. Times were recorded using a Seiko split memory timing instrument and lap times were given to all subjects during the run.

Statistical analysis

Means (SD) were calculated for all tested variables by standard methods. A one-way analysis of variance (ANOVA) was used to determine whether any differences existed between the experimental and control groups on pre-test-dependent measures prior to the training period. At the completion of the experiment, the data were examined for normality of distribution and a two (group) by two (test occasion) ANOVA with repeated measures on the tests factor was performed on the data collected for each test occasion. Follow-up univariate contrasts (Bonferroni) were performed on significant effects to determine the significance of the pairwise comparisons or contrasts. In order to further investigate the pre- to post-training data, Pearson's correlation was performed to analyse the relationship between the changes in MTS and performance and RE. Statistical significance was set at $P < 0.05$.

Results

Analysis of the pre-test data showed a trend suggesting superior 3-km performance in the 3-km test for the C group (see Table 2). Despite this, the pre-training

ANOVA revealed no significant differences between the two groups on any dependent variables. The implications of this pre-test trend in running performance are addressed further in the discussion.

Running and jumping performance

The CMJ height and the 5BT results for the E group improved by 13.2% and 7.8% ($P < 0.01$) respectively, indicating the effectiveness of the plyometric training programme in the current study. The 3-km run time for the E group improved by 2.7% ($P < 0.02$) when comparing the pre- and post-data, equating to an improved mean time of 16.6 s. No significant changes were recorded for the C group on any variable (See Table 2).

RE, $\dot{V}O_{2\max}$, Th_{la} tests

No significant changes were observed in $\dot{V}O_{2\max}$ for the E group. However, following the 6-week training period, RE improved in the E group for each of the tested velocities. The mean increases were 6.7% for 12 km h⁻¹ ($P < 0.004$); 6.4% for 14 km h⁻¹ ($P < 0.002$); and 4.1% for 16 km h⁻¹ ($P < 0.009$, Fig. 2). No significant changes in lactate accumulation and Th_{la} were observed in the E group (Fig. 3). No significant changes were recorded for the C group in any of these tests.

MTS, $F_{iso,\max}$, and RFD tests

The changes in MTS, $F_{iso,\max}$, and RFD over the course of the training period are outlined in Table 3. Whilst MTS was assessed at three different percentages of maximal isometric force (25, 50, and 75%), it was at the heaviest load that the most significant changes occurred. An increase of 14.9% for the left leg and 10.9% ($P < 0.05$) for the right leg were recorded for the MTS test at the 75% load for the E group. The E group recorded improvements in the $F_{iso,\max}$ test of 11.4% ($P < 0.003$) for the left leg and 13.6% ($P < 0.02$) for the

Table 2 Body mass (BM), counter movement jump (CMJ) height, 5-bound test (5BT), lactate threshold (Th_{la}), running economy (RE), $\dot{V}O_{2\max}$, and 3-km performance before and after training. Values are means (SD)

Variable	Experimental group		Control group	
	(n = 8)		(n = 9)	
	Pre	Post	Pre	Post
BM (kg)	74.74 (2.94)	74.80 (2.85)	70.24 (6.47)	70.05 (6.65)
CMJ (m)	0.38 (0.06)	0.43 (0.08)*	0.33 (0.04)	0.32 (0.06)
5BT (m)	10.46 (0.76)	11.28 (0.73)*	10.23 (0.55)	10.33 (0.58)
Th_{la} (mmol l ⁻¹)	4.26 (1.18)	4.03 (1.42)	3.73 (1.08)	4.10 (0.73)
RE (ml kg ⁻¹ min ⁻¹)				
12 km h ⁻¹	26.05 (4.11)	24.30 (3.68)*	24.08 (2.87)	24.21 (3.37)
14 km h ⁻¹	33.35 (5.15)	31.23 (4.27)*	30.62 (3.29)	30.46 (3.98)
16 km h ⁻¹	41.96 (6.14)	40.22 (5.43)*	38.64 (4.95)	38.85 (5.33)
$\dot{V}O_{2\max}$ (ml kg ⁻¹ min ⁻¹)	57.6 (7.7)	59.5 (8.1)	57.8 (5.4)	61.5 (5.9)
3-km time (min)	10.28 (1.26)	10.12 (1.15)*	9.36 (0.57)	9.31 (0.52)

*Significantly different from pre-training value ($P < 0.05$)

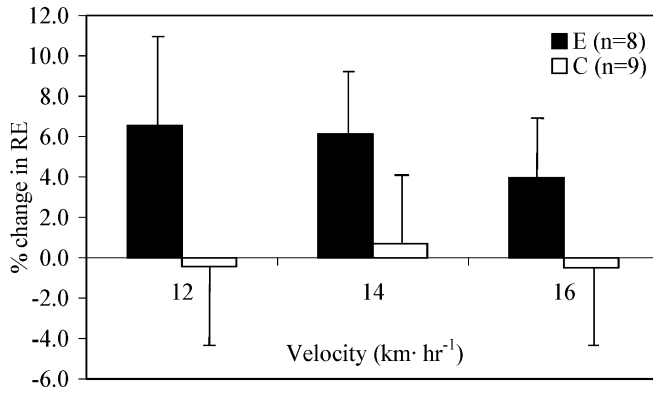


Fig. 2 Percentage (%) changes in running economy (RE) before and after 6 weeks for experimental (E) and control (C) groups

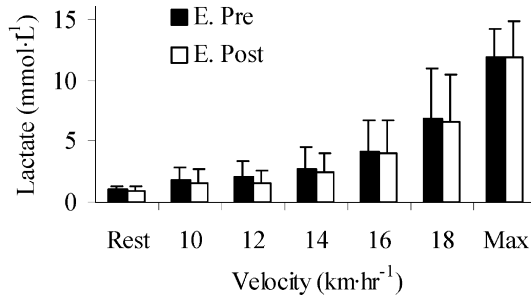


Fig. 3 Pre- and post-training lactate accumulation for the E group ($n=8$)

right leg. RFD also showed improved trends in both legs for the E group, with increases of 14.1% and 15.0% for left and right legs respectively; however these changes were not statistically significant. No significant changes were recorded for the C group in any MTS measures (Table 3).

Correlations between changes in training data

Pearson's correlations were performed between changes in the MTS, RE, jumps, and 3-km test data. These results are presented in Table 4. Significant (though moderate) negative correlations are seen between many of the changes in stiffness variables and change in RE at each velocity. In addition, significant correlations were noted between the change in the 5BT and changes in RE at all velocities. No correlation was found between the changes in RE and change in 3-km running performance.

Discussion

Jump performance

The significant improvements in the 5BT and CMJ height for the E group can be attributed to the

Table 3 Musculotendinous stiffness (MTS), maximal isometric force ($F_{iso,max}$), and rate of force development (RFD) values before and after training. Values are means (SD)

Variable	Experimental group ($n=8$)				Control group ($n=9$)			
	Left leg		Right leg		Left leg		Right leg	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post
MTS ($N\ m^{-1}$)	17,346 (4,689)	17,631 (2,764)	16,992 (3,951)	17,452 (2,969)	15,548 (4,289)	16,337 (2,6350)	16,764 (4,790)	17,014 (2,494)
at 25% $F_{iso,max}$								
MTS ($N\ m^{-1}$)	29,885 (6,180)	33,935 (5,247)	29,666 (7,173)	31,544 (6,080)	27,440 (5,952)	29,203 (5,034)	29,818 (6,065)	32,679 (4,597)
at 50% $F_{iso,max}$								
MTS ($N\ m^{-1}$)	39,935 (7,545)*	45,888 (7,908)*	39,719 (10,865)*	44,058 (8,044)*	37,480 (6,623)	42,197 (5,798)	39,445 (6,3860)	42,415 (5,275)
at 75% $F_{iso,max}$								
$F_{iso,max}$ (N)	1,201 (178)*	1,338 (117)*	1,157 (215)*	1,314 (166)*	1,134 (289)	1,120 (245)	1,196 (212)	1,169 (208)
RFD ($N\ s^{-1}$)	3,364 (1,568)	3,839 (1,125)	2,859 (1,245)	3,289 (1,684)	2,980 (1,332)	2,828 (864)	3,380 (1,062)	2,753 (1,000)

*Significantly different from pre-training value ($P < 0.05$)

Table 4 Correlations between pre- and post-changes (Δ) in MTS, RE, 3-km performance, CMJ and 5BT ($n = 17$)

	Δ RE12	Δ RE14	Δ RE16	Δ 3-km
Δ L25%MTS	-0.28	-0.08	-0.24	0.15
Δ R25%MTS	-0.51*	-0.59**	-0.54*	-0.11
Δ L50%MTS	-0.65**	-0.52*	-0.52*	-0.25
Δ R50%MTS	-0.44*	-0.44*	-0.51*	-0.27
Δ L75%MTS	-0.18	-0.17	-0.46*	0.02
Δ R75%MTS	-0.44*	-0.54*	-0.29	0.01
Δ CMJ	-0.36	-0.22	-0.33	-0.02
Δ 5BT	-0.53*	-0.44*	-0.44*	-0.18
Δ 3-km	0.30	0.23	-0.06	–

* $P < 0.05$, ** $P < 0.01$

plyometric training programme (Table 1). The increases of 7.8% and 13.2% for the 5BT and CMJ tests, respectively, are comparable to those of previous research that has utilised similar training modalities (Wilson et al. 1993; Paavolainen et al. 1999). The fact that these tests represent two distinct entities in terms of both the speed of contraction and ground contact time for SSC movements suggests that the programme was well developed in enhancing a large range of performance tasks.

Plyometric training and running performance

The major aim of the current study was to investigate the mechanism by which plyometric training improved RE and potential running performance. The results showed that 3-km running performance was significantly improved in the E group as a result of a 6-week plyometric training regime. The 2.7% running performance improvement was similar to the approximate increase of 3% reported by Paavolainen et al. (1999) for improved 5-km performance after 9 weeks of simultaneous explosive-strength and endurance training. Furthermore, in the current study RE was significantly improved for the E group at three velocities (Fig. 2). While there was no relationship between changes in RE and 3-km running performance (see Table 4), importantly, the change in 3-km performance was achieved without changes in $\dot{V}O_{2\max}$ and Th_{la} , while body mass showed no change from pre- to post-testing. Therefore, in accordance with the results of Paavolainen et al. (1999), we speculate that it was the change in RE that lead to improved 3-km running performance. This finding is in agreement with previous research that RE is one of the best running

performance indicators (Conley et al. 1984; Noakes 1991; Daniels 1998).

It should be noted, however, that a limitation of the current study was the trend toward differences between the two groups on certain aspects of the pre-test data (see Table 2). Whilst there was no statistical significance, care should be taken when interpreting the results as there was a trend towards the C group having a better pre-test 3-km performance. The clear question that arises is whether it was only the less able runners that exhibited improvements resulting from the plyometric training? However, individual data from the E group demonstrates clearly that improvements in RE and performance were comparable in both faster and slower runners (see Table 5).

Similar to previous research, it was only at the maximal tested load that stiffness levels demonstrated significant changes from the pre- to post-testing occasions for the E group (Wilson et al. 1992). Shorten (1987) suggested that muscle tension appears to be the major determinant of SEC stiffness at low loads, however, as load increases, the stiffness of the SEC approaches that of the tendon, right up to maximum load, in which the stiffness of the SEC equates to the stiffness of the tendon. Therefore, we believe it is reasonable to assume that because MTS only increased in the maximal load condition between the two test occasions, then the adaptations of the plyometric training occurred by primarily increasing the stiffness of the tendon as opposed to increases in other structures.

The findings of the current study, that plyometric training increased MTS leading to improved RE (and jump performance), are strongly supported by a number of recent studies that have examined the relationship between leg stiffness and performance in both sprinting and continuous running. Heise and Martin (1998) reported an inverse relationship between vertical leg stiffness and RE in 16 recreational runners. The authors concluded that less economical runners possessed a running style that was more compliant during ground contact. Similarly, Dalleau et al. (1998) noted that the energy cost of treadmill running was significantly related to the stiffness of the propulsive leg. Furthermore, Chelly and Denis (2001) found that leg stiffness assessed during a hopping task was significantly correlated to maximal sprint velocity in handball players. It was stated by these authors that leg stiffness is a good indicator of an individual's reactive power, i.e. their ability

Table 5 Individual 3-km time, RE, and $\dot{V}O_{2\max}$ before and after 6 weeks plyometric training for three of the fastest runners (3-km time) in the experimental group. Data clearly demonstrates improvements in running performance and RE

Variable	Subject 1		Subject 2		Subject 3	
	Pre	Post	Pre	Post	Pre	Post
3-km time (min)	9.19	8.59	9.23	9.13	9.35	9.26
RE (ml $kg^{-1} min^{-1}$)						
12 km h^{-1}	24.18	21.00	22.08	20.05	25.03	24.63
14 km h^{-1}	32.33	28.60	26.60	25.67	33.75	31.47
16 km h^{-1}	38.23	37.3	34.57	32.47	44.63	41.88
$\dot{V}O_{2\max}$ (ml $kg^{-1} min^{-1}$)	57.97	61.40	60.30	61.45	70.77	71.32

to absorb and restitute power during forceful SSC actions (such as running). Therefore, we believe that in the current study, the change in MTS due to plyometric training increased reactive power such that the energy cost of running was significantly reduced. Providing some support for this was the finding of a number of significant correlations between change in MTS and change in RE as a result of the plyometric training (see Table 4). Conceivably, such adaptations could have resulted in changes to either stride length or stride frequency, due to the athletes achieving greater forward propulsion per foot strike at a decreased energy cost. In addition, the correlation between change in 5BT, a test of horizontal propulsion, and change in RE (Table 4) supports these findings. Such a mechanism also accounts for the significant increase in 5BT performance experienced by the E group after plyometric training. However, changes in stride length or stride frequency being responsible for the above mentioned adaptation remains as speculation, since changes in stride length and stride frequency were not measured in the current study.

It is important to note that different methodologies have been used in the literature to measure stiffness. Previous research has calculated total leg stiffness during hopping or running tasks (Dalleau et al. 1988; Chelly and Denis 2001). In the current study stiffness was measured unilaterally in a protocol designed to isolate ankle stiffness. The relationship between total leg stiffness and isolated lower limb stiffness is currently unclear. However, Farley et al. (1998) reported that adjustments to ankle stiffness are most important to adjustments in total leg stiffness. The reader is referred to Farley et al. (1998) for a more detailed analysis of leg and joint stiffness interactions.

In conclusion, we have shown that a 6-week plyometric training programme led to improvements in RE. The mechanism underlying this change appears to be a result of an increase in lower body MTS with plyometric training leading to improved reactive power and a decreased energy cost of running. We further speculate that the improved RE led to changes in 3-km running performance, particularly as there were no corresponding alterations in $\dot{V}O_{2\max}$ or Th_{la} . Further research is needed to establish whether such running improvements derive from an increased stride length or stride frequency, or a combination of the two. Such findings have implications for distance runners who do not routinely perform progressive plyometric training, even though running itself is an intense SSC activity.

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